

Problem

What kind of filter can be used to select a signal of one particular radio station?

- (a) low-pass
- (b) high-pass
- (c) band-pass
- (d) band-stop

Step-by-step solution**Step 1 of 1**

A band-pass filter passes frequencies within a frequency band and blocks or attenuates frequencies outside the band. A band-pass filter best suited to select a signal of one particular radio station.

Therefore the correct option is **(c)**.

Low pass filter passes low frequencies and stops high frequencies. So normally low pass filter is not best suited to select range of high frequencies. Thus option **(a)** is wrong.

High pass filter passes high frequencies and stop low frequencies. So high pass filter not suited to select range of low frequencies. So option **(b)** is wrong.

Band stop filter passes frequencies outside a frequency band and blocks or attenuates frequencies within the band. Thus band stop filter best suit for rejection not for selection. So option **(d)** is wrong.